

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Final
Semester: Fall 2023

Duration: 1 Hour 45 Min
Full Marks: 35

CSE 470: Software Engineering

SET A

Name:

ID:

Section:

Q1. Imagine a scenario where a research organization is embarking on a complex project to develop an information system to support their research activities that will involve researchers from all over the world, collaborating with multiple country students and researchers into the system. The requirements for the system are not fully known upfront, and they may evolve as the organization gains a better understanding of its needs as the project progresses. This involves risk, as any invasion of the research data can create an international problem. The organization is willing to take time and build a team that can deliver the project while delivering reports on problems and complexity that may arise. One of the engineers suggested using the Waterfall model for the above scenario.

- a. State whether the suggestion is valid for the above scenario or not and **why**. [2] [CO1]
- b. **What** model do you suggest, if not waterfall, and **why**? [3] [CO1]

Q2. Now that BracU has shifted to its new campus, the IT department is creating a department-level printing service where all the printers in a department are connected to a single server. All print requests are sent to the server, and the server maintains a printing queue and selects a printer that is close to the faculty and is free. You have to write the code for the server that maintains this print queue.

- a) What design pattern should be implemented? Justify **Why**? [4] [CO3]
- b) Write the code (in any programming/scripting language) for the design pattern. [3.5] [CO3]

Q3. Consider the following code snippet. [CO4]

- a. **Determine the Specialization Index (SIX) for the Sports Class.** [5]
- b. **Illustrate** the Control Flow Graph (CFG) for the hostClub() method, and remember to label the relevant node numbers on your question paper. [7]
- c. **Calculate** the cyclomatic complexity of the control flow graph drawn for the hostClub() method. [3]

<pre> public class Events { public void startEvents() { System.out.println("BRAC University events have started."); } public void endEvents() { System.out.println("BRAC University events have ended."); } private void displayWinner() { System.out.println("Winner of the BRAC University events is not yet decided."); } } public class Sports extends Events { private void displayWinner() { System.out.println("Winner of the BRAC University Sports event is the Red Team."); } public void award() { System.out.println("The Most Valuable Player of BRAC University Sports is Hasan."); } } public class NewCampusEvents extends Sports { public void award() { System.out.println("The Most Valuable Player of BRAC University New Campus Events is Habib."); } } </pre>	<pre> public void hostClub() { boolean clubHosted = true; int year = 2023; boolean oca = true; boolean other = false; for (int i = 0; i < 3; i++) { for (int j = 0; j < 3; j++) { if (i == 2) { oca = true; } if (j == 2) { oca = true; } if (i == 1) { other = true; } if (j == 1) { other = true; } } } public void starStudentAward() { System.out.println("The Star Student of BRAC University New Campus is Mita"); } } </pre>
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Q4. Consider the following below code snippet.

[CO5]

- Identify** at-least 3 code smells from the below code and name them. [3]
- Refractor** the source code to remove 4 different Code smells (write the code segment in your copy which has code smell and then show the correct way to refactor it). [4.5]

NB: The “**System.out.println**” statements should not cause any smells.

```

private static final int count_variable
public class ShoppingSystem {
    public static void main(String[] args) {
        // Creating items
        Item_name="laptop"
        Item_price=1200
        Item laptop = new Item(Item_name, Item_price);
        Item_name="headphone"
        Item_price=80
        Item headphones = new Item(Item_name, Item_price);

        // Creating a customer object
        Customer customer = new Customer("John Doe", true);

        // Creating a shopping cart object
        ShoppingCart shoppingCart = new ShoppingCart(customer);

        // Adding items to the shopping cart
        shoppingCart.addItem(laptop);
        shoppingCart.addItem(headphones);
        shoppingCart.addItem(mouse);

        // Calculating and displaying the total cost
        double totalCost = 0.0;

        for (Item item : shoppingCart.items) {
            //getting the price value
            double itemPrice = item.gtPr();
            double discountRate = 0.0;
            discountRate = customer.isMember() ? 0.2 : 0.1;
            //Getting Discount of 0.2 if member

            if (itemPrice > 100) {
                totalCost += itemPrice - (itemPrice *
discountRate);
            }
            else
            {
                totalCost += itemPrice;
            }
        }

        System.out.println("Total cost for " +
customer.gtNm() + ": $" + totalCost);
        //Getting name and printing total cost
    }
}

```

```

class Customer {
    private String name;
    private boolean isMember;

    public Customer(String name, boolean
isMember) {
        this.name = name;
        this.isMember = isMember;
    }

    public String gtNm() {
        return name;
    }

    public boolean isMember() {
        return isMember;
    }
}

class Item {
    private String name;
    private double price;

    public Item(String nm, double pe) {
        this.name = nm;
        this.price = pe;
    }

    public double gtPr() {
        return price;
    }
}

class ShoppingCart {
    private Customer customer;
    public List<Item> items;
    public ShoppingCart(Customer
customer) {
        this.customer = customer;
        this.items = new ArrayList<>();
    }
}

```

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